

Vuln Spectra

Functionality & Project Report

1. Project Description

Vuln Spectra is a web-based penetration testing and vulnerability assessment platform designed specifically for local websites, small businesses, and independent developers. It enables users to scan and analyze their web applications using real-world Kali Linux tools in a simplified and accessible interface.

2. About the Project

This platform is built to bridge the gap between professional cybersecurity tools and non-technical users. Many small organizations and developers lack the resources or expertise to perform security testing. Vuln Spectra provides an affordable and easy-to-use solution to test website security effectively.

3. Problem Statement

Cyber attacks in Pakistan are increasing rapidly, especially targeting small and local websites. Many businesses suffer from vulnerabilities due to lack of awareness, limited financial resources, and absence of proper security tools. Several local websites have faced defacement attacks, data leaks, and unauthorized access due to weak security practices.

For example, various government and educational websites in Pakistan have been defaced by hackers in past years, highlighting the urgent need for accessible security solutions.

4. Proposed Solution

Vuln Spectra provides an automated vulnerability scanning platform that uses real penetration testing tools like **Nmap**, **Nikto**, **SQLMap**, and **SSLScan**. It simplifies complex security analysis and presents results in both technical and non-technical formats using AI-generated reports.

5. System Workflow

1. **User Authentication:** The user signs up or logs into the system.
2. **Dashboard Access:** After authentication, the user accesses the main dashboard.
3. **Scan Input:** User enters the target URL.
4. **Scan Mode Selection:**
 - **Auto Scan:** Runs all tools automatically for beginners.
 - **Manual Scan:** Allows advanced users to select specific tools.

- 5. **Scanning Process:** The system executes real Kali Linux tools on the backend.
- 6. **Results Display:**
 - Vulnerabilities detected
 - Severity levels
- 7. **AI Report Generation:**
 - Explains vulnerabilities in simple language
 - Provides remediation steps
- 8. **Raw Output:**
 - Technical users can view original tool outputs

6. Key Features

- Automated vulnerability scanning
- AI-based report generation
- Step-by-step patching guidance
- Manual and auto scan modes
- Raw output for technical users
- User-friendly dashboard
- Real Kali Linux tools integration

7. Technology Stack

Frontend

Technology
Vite
React
TypeScript
Deployed on Vercel

Backend

Technology
Node.js (Express)
JavaScript
Deployed on AWS (Ubuntu Server)
Integration with Linux-based security tools

8. Conclusion

Vuln Spectra is a powerful and practical solution for improving web security in environments where resources are limited. By combining real security tools with AI-driven reporting, it empowers both technical and non-technical users to identify and fix vulnerabilities effectively.